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23SPJ607: Socially Relevant Project

Collaborative GIS Based Road Editing System

GUIDE

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PROBLEM STATEMENT

“Maintaining accurate rural road networks remains critically inefficient due to the reliance on labor-intensive manual digitization and the lack of an integrated GIS platform that combines AI-driven automation with real-time, conflict-free collaborative editing.”

MOTIVATION

- Emergency response systems
- Manual Systems are inefficient
- Intuitive and Accessible
- Social Impact
- Current maps tend to induce accidents

LITERATURE REVIEW

Collaborative Workflow

- Jing, C et.al (2019)
- "A Lightweight Collaborative GIS Data Editing Approach to Support Urban Planning."

Web-based

- Butt, Mahmood et.al (2018)
- "GeoWebEX: an open-source online system for synchronous collaboration on geographic information."

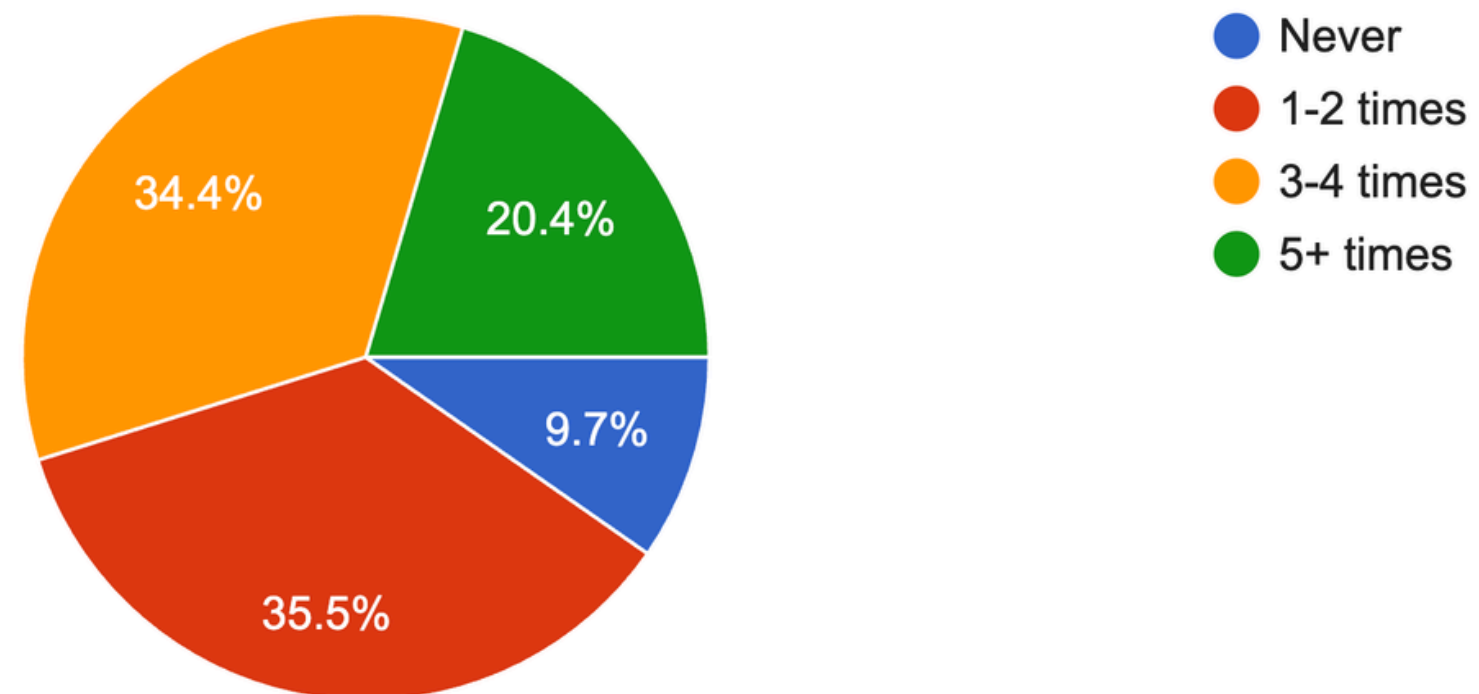
Machine assistance

- Bastani et.al (2019)
- "Machine-Assisted Map Editing."

BACKGROUND STUDY - SURVEY

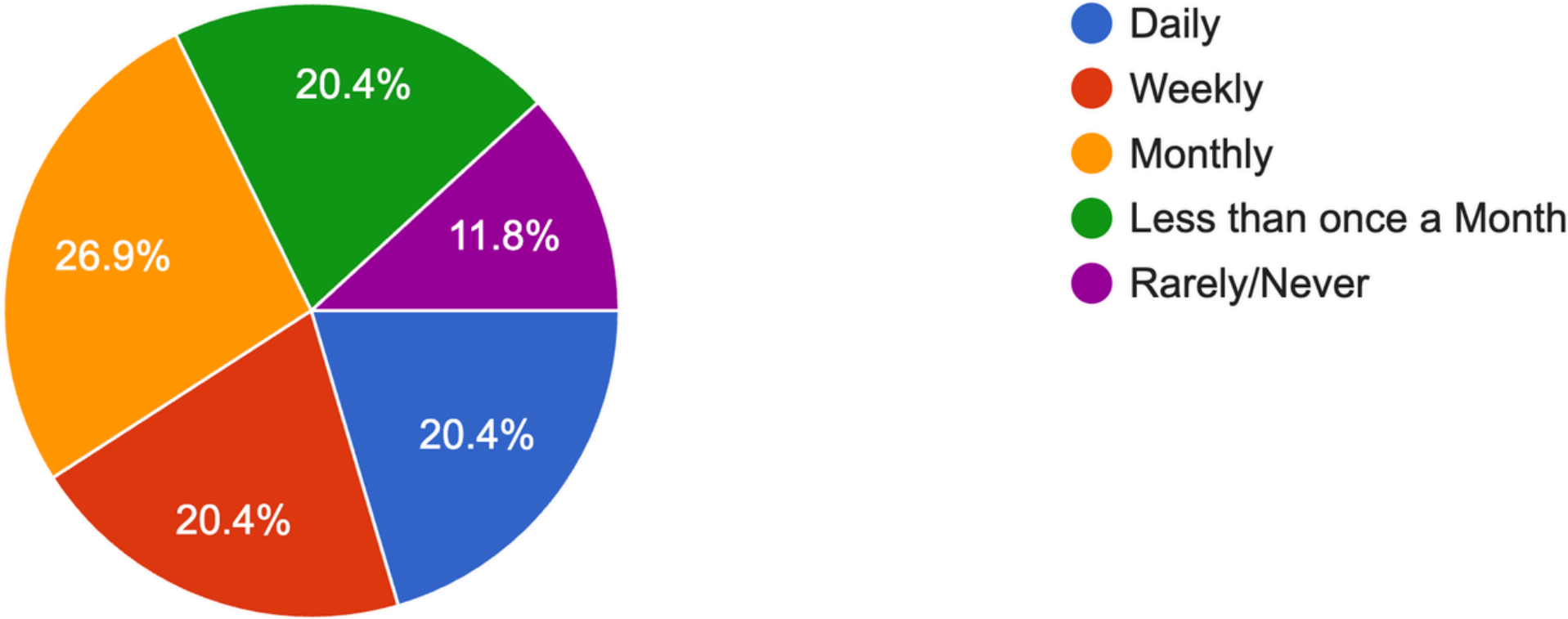
Have you ever been misled by a map in Kerala, or ended up in a dead end, private property, or a narrow road?

93 responses



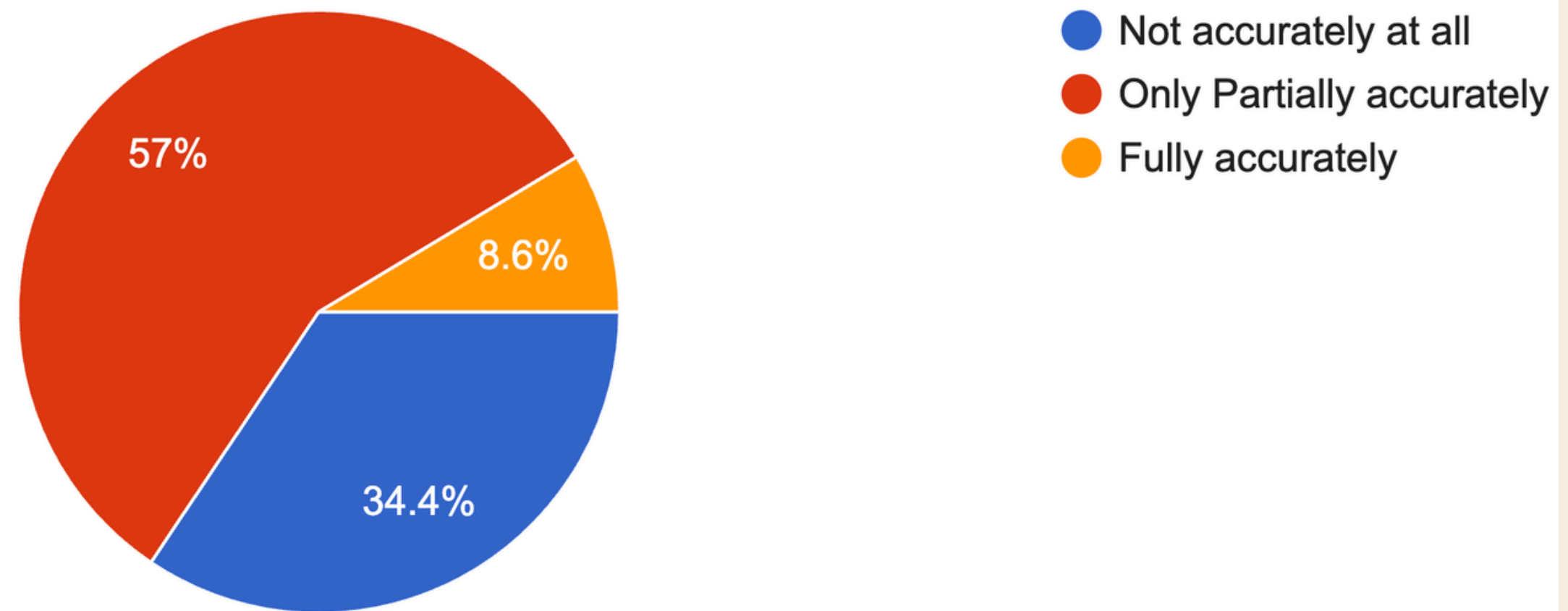
How often do you have to guide Swiggy/Amazon or cab drivers over the phone because your exact lane or home location is missing/incorrect on the map?

93 responses



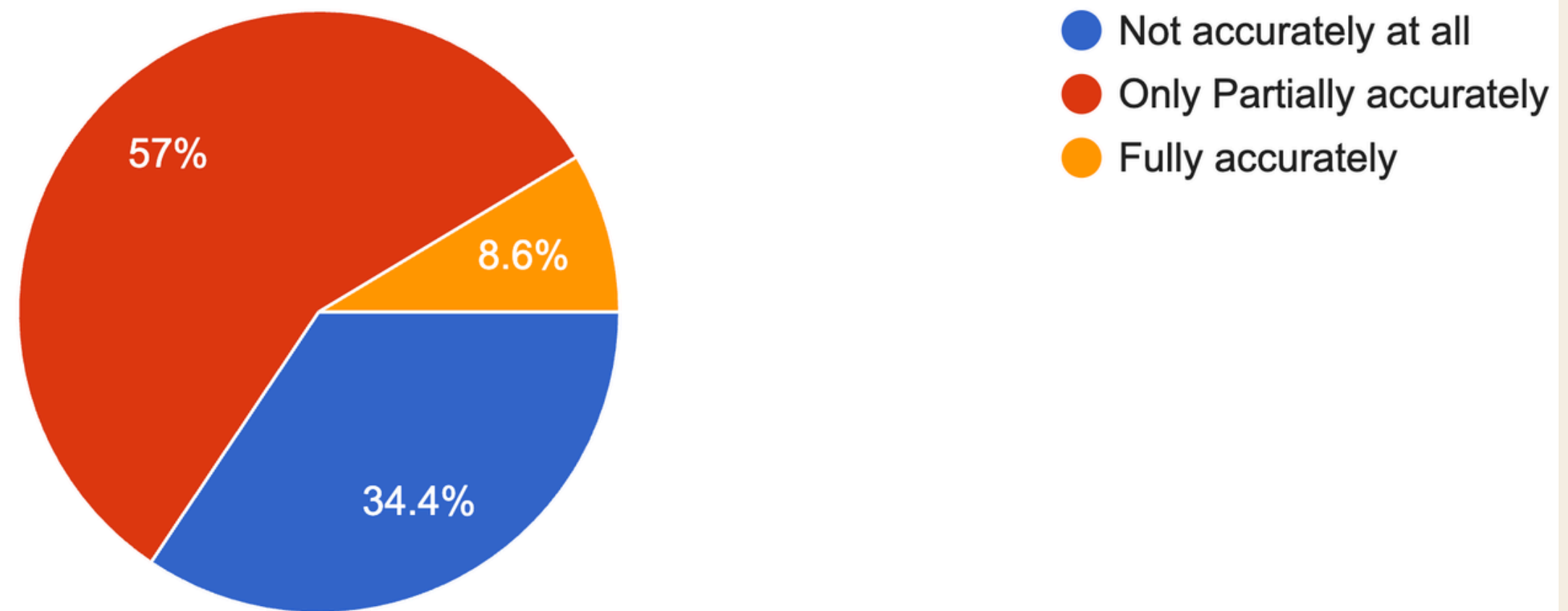
Thinking about your campus or hometown, how accurately are the small inner pathways, local bypasses, and shortcuts shown on Google Maps?

93 responses



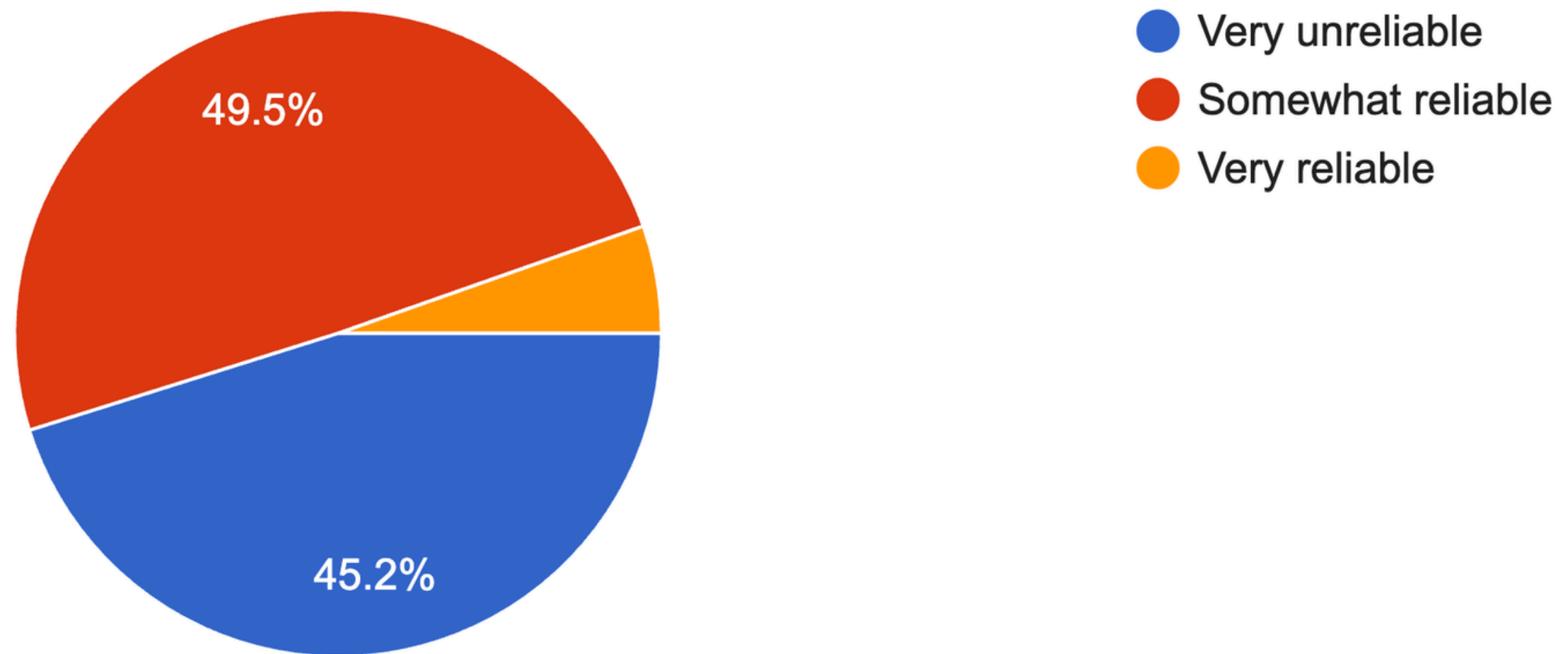
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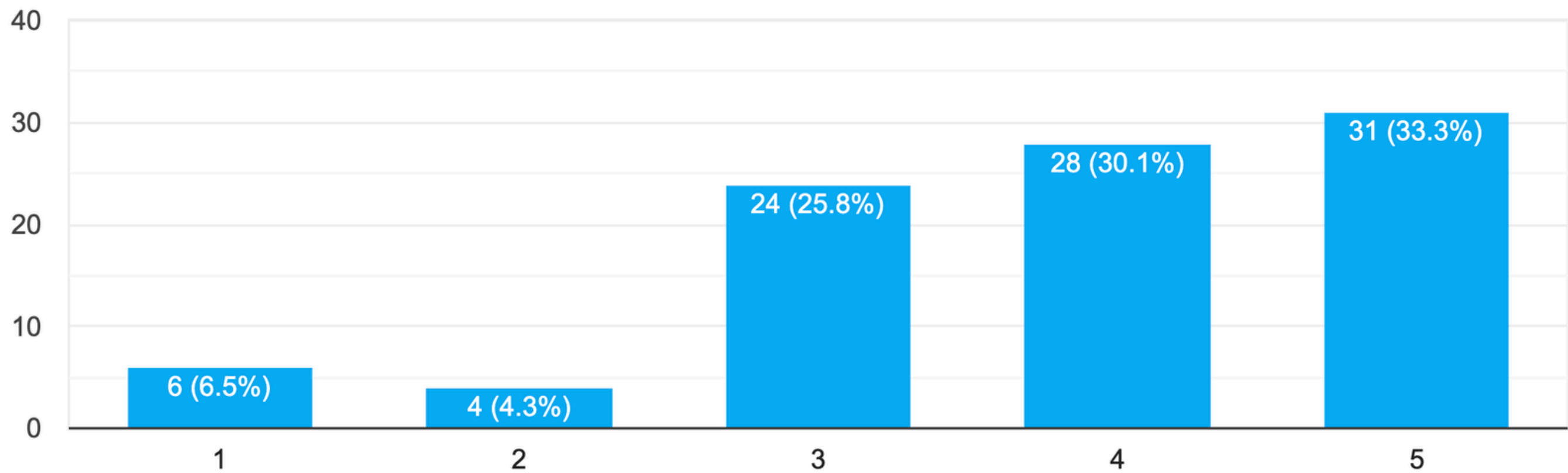
During Kerala monsoons or floods, how reliable are standard digital maps for finding safe, alternative routes (avoiding waterlogged roads or landslides)?

93 responses



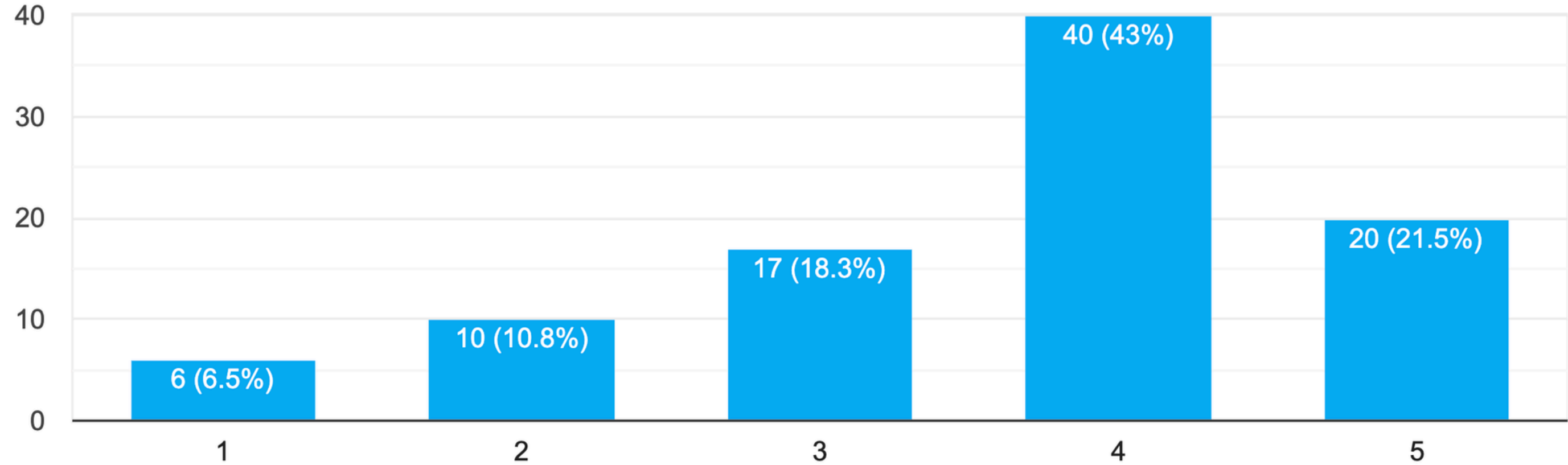
Rate the level of frustration you feel when digital maps fail you in your locality (1 = No frustration, 5 = Extreme frustration).

93 responses



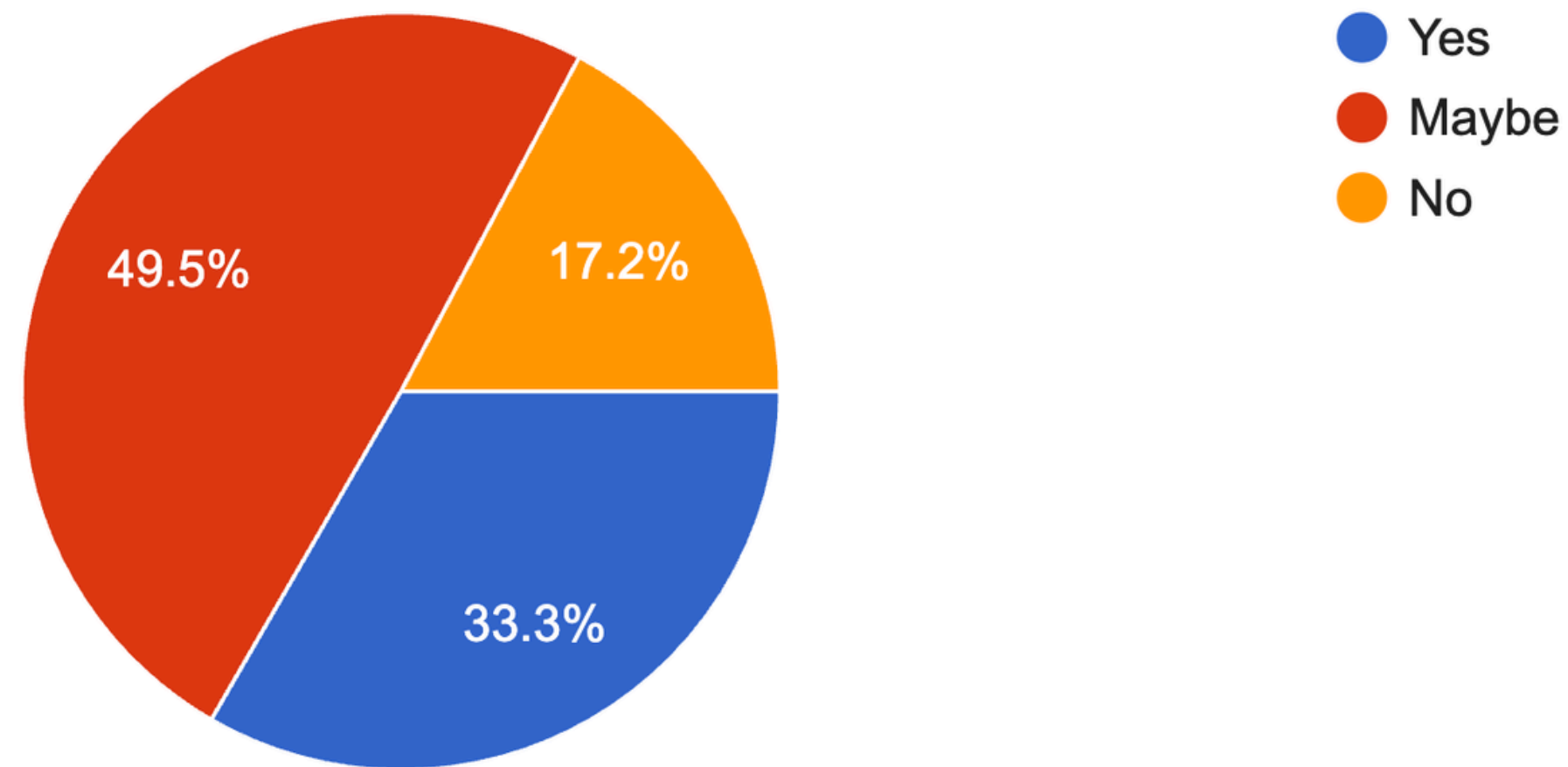
If you had to map your locality for a group project, how useful would a 'Google Docs-style' real-time collaborative map editor be?

93 responses



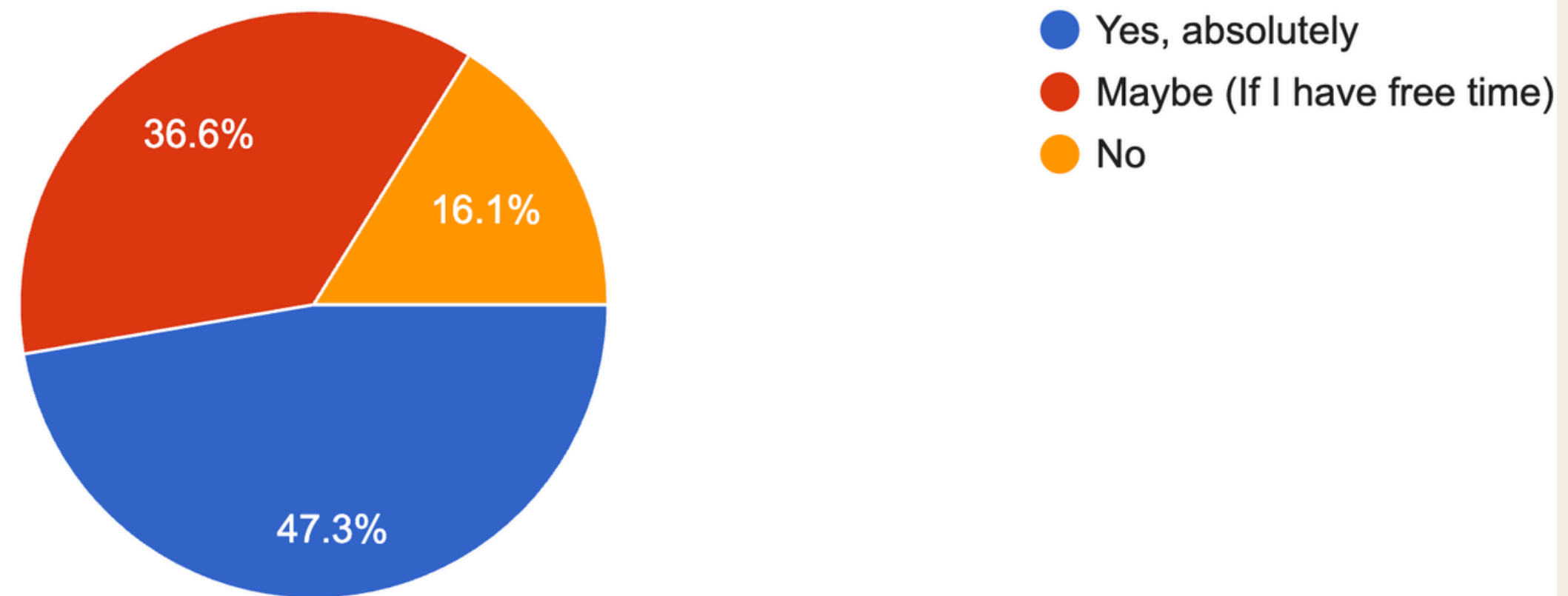
Do you believe that using AI to automatically draw 70% of a map—leaving only quick corrections (verification) to humans—is better than drawing it manually from scratch?

93 responses

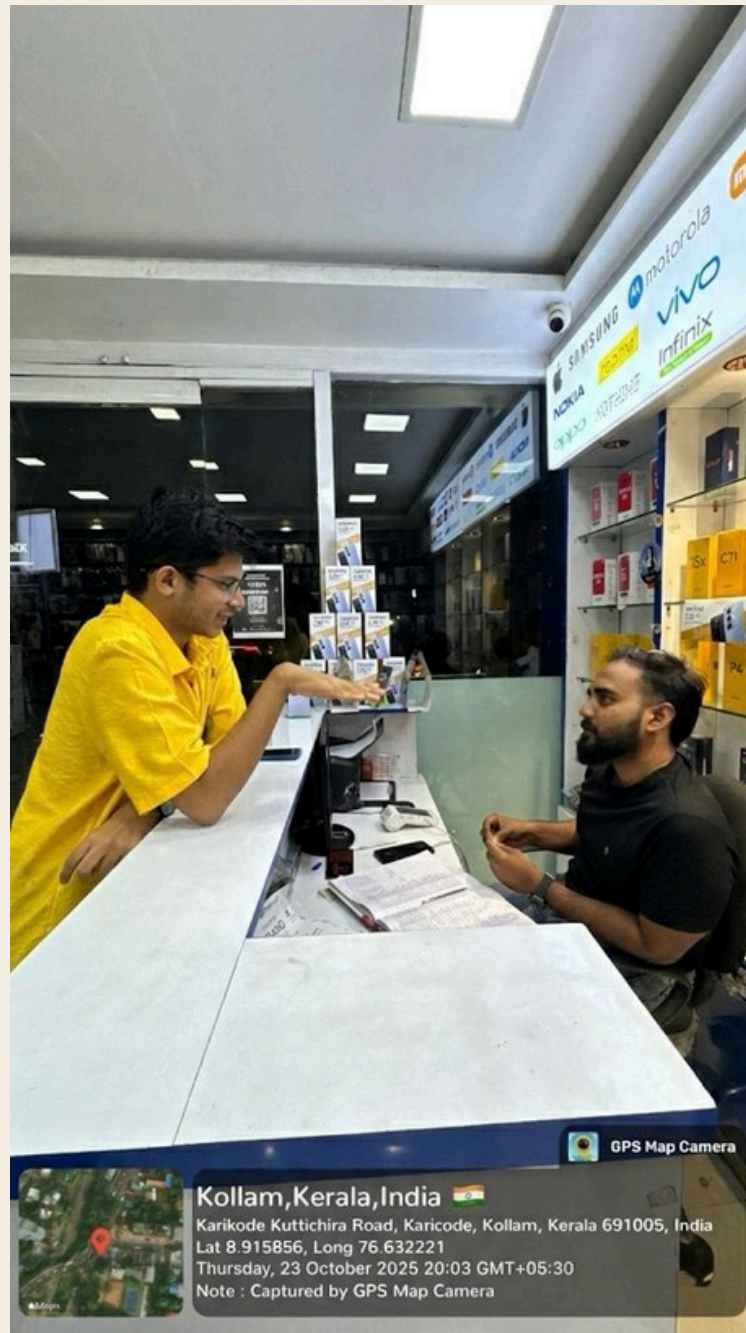


If an easy web platform existed, would you volunteer 10 minutes to verify and correct AI-generated maps of your own neighborhood to help local students, tourists, and emergency responders?

93 responses



FIELD WORK



PROFESSIONALS WE INTERACTED WITH

To understand:

- How current map editing platforms operate
- Challenges in real-world GIS workflows
- Limitations of existing collaborative systems
- Scope of AI-assisted mapping in industry



Dr.Suresh Francis
Scientist, KSREC



Kiran Raj
Software Programmer, KSREC

OBJECTIVES

- Develop web-based collaborative map editor (Next.js + Leaflet)
- Enable precise MultiLineString road editing
- Integrate AI-based road extraction (U-Net model)
- Implement real-time collaboration (WebSockets)
- Add road-lock mechanism
- Store and manage spatial data using PostGIS

PROPOSED SYSTEM

- Next.js-based web application with server-side routing.
- Leaflet OSM map with road visualization + editing tools.
- AI model trained for automatic road extraction from images.
- Real-time operations powered by WebSockets.
- PostGIS used for spatial queries, storing roads, and indexing.
- Road-locking to reserve roads during collaborative edits.

TECHNOLOGIES

Frontend

- Next.js
- OSM Viewer

Backend

- Next.js API Routes
- Geoserver to publish changes immediately to users
- AI Model API
- WebSocket real-time sync layer

Database

- PostgreSQL + PostGIS
- Spatial indexing (GiST)
- Store roads, boundaries

SYSTEM DEVELOPMENT

FRONTEND DEVELOPMENT

- Leaflet layers for roads & village polygons
- Editable nodes for MultiLineString geometry
- Drag-and-drop road editing UI
- Road-lock display (highlight locked roads)

SYSTEM DEVELOPMENT

BACKEND DEVELOPMENT

- Next.js API for CRUD operations
- Geoserver connected to the database
- WebSocket events

AI MODEL DEVELOPMENT

- Collect & prepare satellite imagery
- Train model for road segmentation
- Convert predictions to GeoJSON

SDGs Achieved

SDG 9: Industry, Innovation, and Infrastructure

SDG 11: Sustainable Cities and Communities

SDG 13: Climate Action

CHALLENGES FACED

- Handling MultiLineString geometry without breaking structure.
- Training an accurate AI model with limited dataset.
- Real-time sync issues when multiple users edit simultaneously.
- Handling PostGIS queries.

CONCLUSION

- Develop a collaborative, GIS-based Road Editing System.
- Support manual and automatic road editing with real-time sync.
- Enhance community participation in map creation.
- Robust tech stack: Next.js, Leaflet, PostGIS, WebSocket, GeoServer

REFERENCES

1. Jing, C., Zhu, Y., Fu, J., & Dong, M. (2019). A Lightweight Collaborative GIS Data Editing Approach to Support Urban Planning.
2. Butt, M. A., Mahmood, S. A., & Raza, S. M. H. (2018). GeoWebEX: an open-source online system for synchronous collaboration on geographic information.
3. Pollino, M., Fattoruso, G., La Porta, L., Della Rocca, A. B., & James, V. (2012). Collaborative Open Source Geospatial Tools and Maps Supporting the Response Planning to Disastrous Earthquake Events.

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THANK YOU

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